

TidyTuesday Week 31

Pranay Gundam

Saturday 1st August, 2026

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to another thrilling week of Randomly Yoking Two Economic Time Series Together And Praying, the world's most rigorous scientific enterprise, if by "rigorous" you mean "a dice roll wearing a lab coat." As always, the usual disclaimer applies before anyone drafts a policy memo off this: these pairs are chosen by a machine with no taste, no agenda, and no idea what any of these acronyms mean. Spurious is the house style here. If something looks meaningful, that's the correlation talking, and it lies.

We opened Sunday on a real barnburner: the World Pandemic Uncertainty Index for New Zealand versus Total Private Construction Spending on Communication in the United States. I know, I know, try to contain yourselves. This one did us the enormous favor of being useless right out of the gate – an R-squared of 0.008 and a p-value of 0.32, meaning even the raw regression, the one that will grab any excuse to look impressed, just shrugged. Detrended, it got even sadder. A rare case of a pairing so unrelated that not even our trend-hungry method could fake a spark. Refreshingly honest failure. Ten out of ten, no notes.

Then came the two showstoppers, and this is where you have to keep your wits about you. Monday paired a Northeast core CPI series with U.S. transportation-services CPI and posted a raw R-squared of 0.97, which is the kind of number that makes people whisper "Nobel." But here's the twist that makes this week actually interesting: it held up detrended too, still wildly significant with an R-squared around 0.84. That's not the usual trend mirage collapsing under scrutiny – that's two inflation measures genuinely moving together even after you strip the shared upward march out, which, given that both are literally chunks of the same consumer price index, is about as close to "well, obviously" as this project ever gets. Tuesday's pairing of an "Other Goods and Services" CPI against a Medical Services price index pulled the exact opposite stunt: a gorgeous 0.98 raw R-squared that evaporated into a 0.00005 nothingburger the second we detrended it. Classic. Two things that both went up over the years, holding hands purely because time passed. Chef's kiss of spuriousness.

The back half of the week was mostly a parade of the mildly desperate. Government employment-and-training benefits versus a discontinued bank loan rate found basically nothing raw, then coughed up a barely-significant detrended p-value of 0.038 with an R-squared of 0.22 – the statistical equivalent of a polite golf clap. Great Lakes financial-services spending versus a circulatory-disease price index looked decent raw and then folded detrended, another trend impostor unmasked. And we closed Saturday on Middle Atlantic residential natural gas prices versus the San Francisco Tech Pulse, which squeaked in at $p=0.046$ detrended with an R-squared of 0.04,

i.e., technically significant and practically a rounding error. So to recap: one genuinely believable relationship (that was two flavors of CPI, so, cheating), a couple of textbook trend mirages, and a handful of coin-flips pretending to matter. Do not build anything on any of this. See you next week for more.

2 Date: 2026-07-27

Series ID: WUPINZL

This series is titled World Pandemic Uncertainty Index for New Zealand and has a frequency of Quarterly. The units are Index and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1996-01-01 and the observation end date is 2026-01-01. The popularity of this series is 1.

Series ID: PRCMUCON

This series is titled Total Private Construction Spending: Communication in the United States and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1993-01-01 and the observation end date is 2026-05-01. The popularity of this series is 1.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_PRCMUCON	R-squared:	0.008			
Model:	OLS	Adj. R-squared:	-0.000			
Method:	Least Squares	F-statistic:	0.9862			
Date:	Mon, 27 Jul 2026	Prob (F-statistic):	0.323			
Time:	14:06:59	Log-Likelihood:	-897.76			
No. Observations:	121	AIC:	1800.			
Df Residuals:	119	BIC:	1805.			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t 	[0.025	0.975]
const	1632.8538	37.497	43.547	0.000	1558.607	1707.101
value_fred_WUPINZL	10.4150	10.487	0.993	0.323	-10.351	31.181
Omnibus:	3.754	Durbin-Watson:	0.365			
Prob(Omnibus):	0.153	Jarque-Bera (JB):	2.215			
Skew:	0.059	Prob(JB):	0.330			
Kurtosis:	2.348	Cond. No.	3.63			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

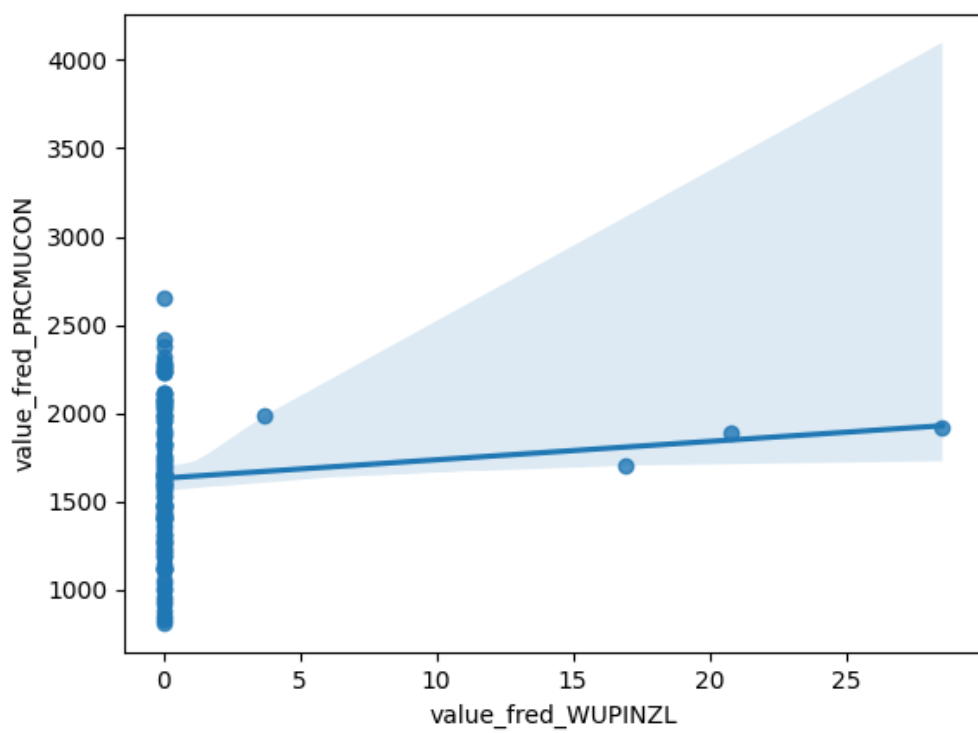


Figure 1: Raw Regression Plot for 2026-07-27

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_PRCMUCON_detrended	R-squared:	0.005
Model:	OLS	Adj. R-squared:	-0.004
Method:	Least Squares	F-statistic:	0.5697
Date:	Mon, 27 Jul 2026	Prob (F-statistic):	0.452
Time:	14:06:59	Log-Likelihood:	-858.73
No. Observations:	121	AIC:	1721.
Df Residuals:	119	BIC:	1727.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.119e-11	26.801	4.17e-13	1.000	-53.069	53.069
value_fred_WUPINZL_detrended	-5.8542	7.756	-0.755	0.452	-21.212	9.503

Omnibus:	14.526	Durbin-Watson:	0.687
Prob(Omnibus):	0.001	Jarque-Bera (JB):	17.629
Skew:	0.701	Prob(JB):	0.000149
Kurtosis:	4.238	Cond. No.	3.46

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

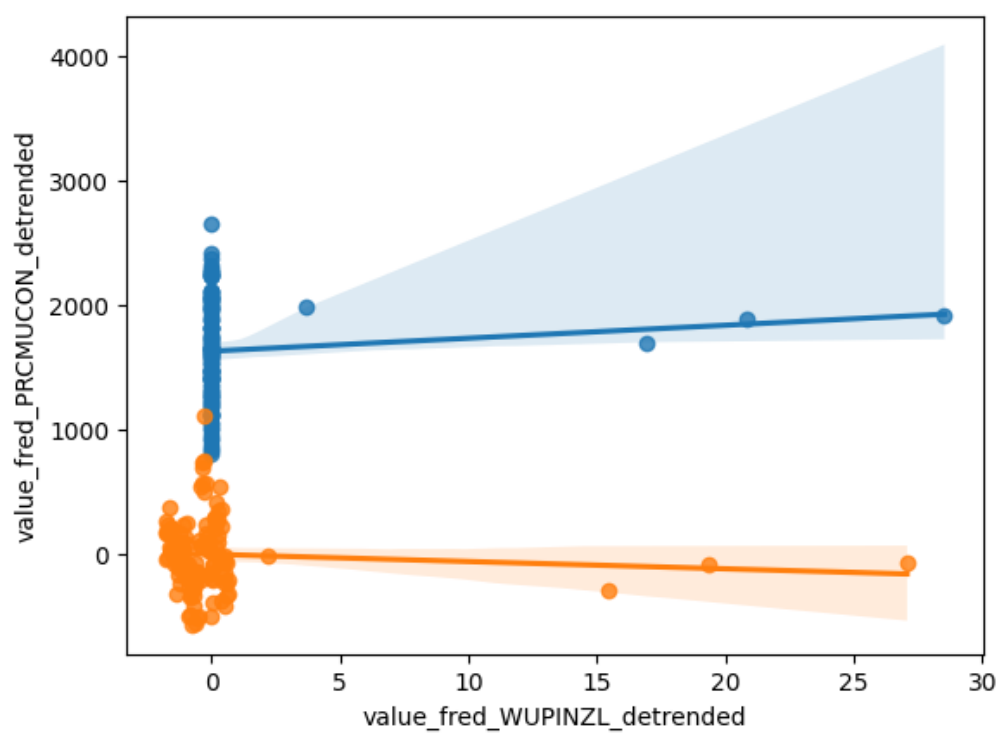


Figure 2: Detrended Regression Plot for 2026-07-27

3 Date: 2026-07-28

Series ID: CUUSX100SA0L1E

This series is titled Consumer Price Index for All Urban Consumers: All Items Less Food and Energy in Northeast - Size Class B/C and has a frequency of Semiannual. The units are Index Dec 1997=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1998-01-01 and the observation end date is 2026-01-01. The popularity of this series is 1.

Series ID: CWSR0000SAS4

This series is titled Consumer Price Index for All Urban Wage Earners and Clerical Workers: Transportation Services in U.S. City Average and has a frequency of Monthly. The units are Index 1982-1984=100 and the seasonal adjustment is Seasonally Adjusted. The observation start date is 1956-01-01 and the observation end date is 2026-06-01. The popularity of this series is 1.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_CWSR0000SAS4	R-squared:	0.974
Model:	OLS	Adj. R-squared:	0.973
Method:	Least Squares	F-statistic:	2037.
Date:	Tue, 28 Jul 2026	Prob (F-statistic):	3.80e-45
Time:	13:25:41	Log-Likelihood:	-227.11
No. Observations:	57	AIC:	458.2
Df Residuals:	55	BIC:	462.3
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-174.8164	10.353	-16.886	0.000	-195.564	-154.069
value_fred_CUUSX100SA0L1E	3.3910	0.075	45.134	0.000	3.240	3.542

Omnibus:	1.760	Durbin-Watson:	0.172
Prob(Omnibus):	0.415	Jarque-Bera (JB):	0.997
Skew:	0.247	Prob(JB):	0.607
Kurtosis:	3.420	Cond. No.	813.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

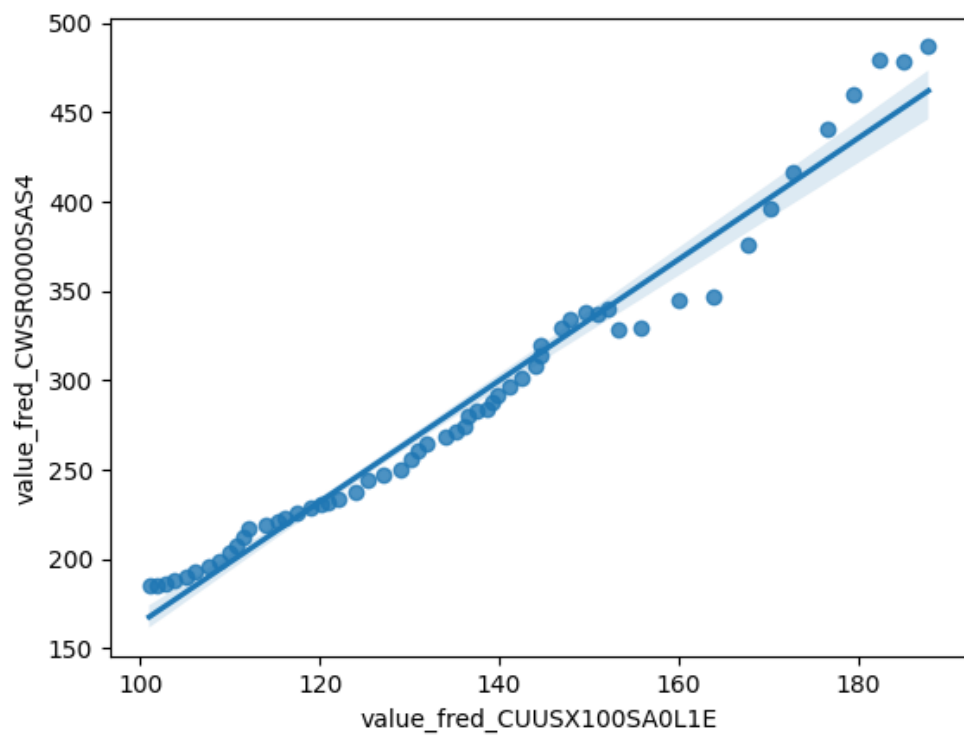


Figure 3: Raw Regression Plot for 2026-07-28

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_CWSR0000SAS4_detrended	R-squared:	0.841
Model:	OLS	Adj. R-squared:	0.838
Method:	Least Squares	F-statistic:	290.1
Date:	Tue, 28 Jul 2026	Prob (F-statistic):	1.35e-23
Time:	13:25:41	Log-Likelihood:	-212.21
No. Observations:	57	AIC:	428.4
Df Residuals:	55	BIC:	432.5
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-1.098e-11	1.350	-8.13e-12	1.000	-2.706	2.706
value_fred_CUUSX100SA0L1E_detrended	5.2535	0.308	17.033	0.000	4.635	5.872

Omnibus:	3.612	Durbin-Watson:	0.292
Prob(Omnibus):	0.164	Jarque-Bera (JB):	2.762
Skew:	-0.345	Prob(JB):	0.251
Kurtosis:	3.828	Cond. No.	4.38

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

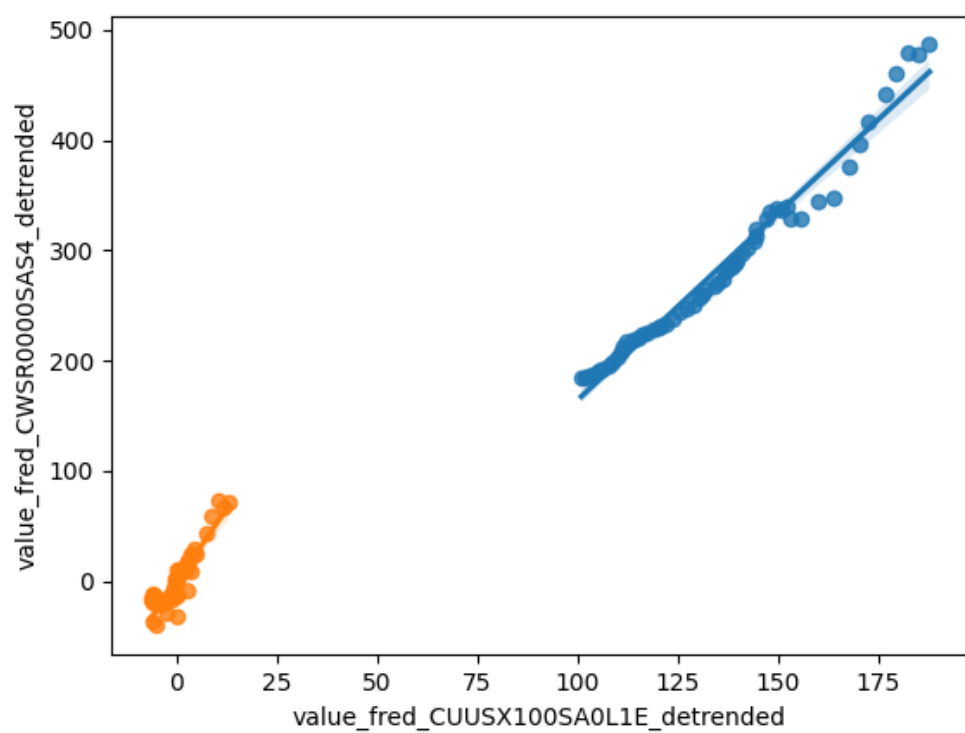


Figure 4: Detrended Regression Plot for 2026-07-28

4 Date: 2026-07-29

Series ID: CWUR0000SAG

This series is titled Consumer Price Index for All Urban Wage Earners and Clerical Workers: Other Goods and Services in U.S. City Average and has a frequency of Monthly. The units are Index 1982-1984=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1967-01-01 and the observation end date is 2026-06-01. The popularity of this series is 2.

Series ID: MESEPRPIHCSA

This series is titled Medical Services Expenditures by Provider Price Index and has a frequency of Annual. The units are Index 2017=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2000-01-01 and the observation end date is 2021-01-01. The popularity of this series is 3.

4.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_MESEPRPIHCSA	R-squared:	0.981			
Model:	OLS	Adj. R-squared:	0.980			
Method:	Least Squares	F-statistic:	1020.			
Date:	Wed, 29 Jul 2026	Prob (F-statistic):	1.23e-18			
Time:	13:34:17	Log-Likelihood:	-48.503			
No. Observations:	22	AIC:	101.0			
Df Residuals:	20	BIC:	103.2			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t 	[0.025	0.975]
const	4.9487	2.574	1.923	0.069	-0.421	10.318
value_fred_CWUR0000SAG	0.2061	0.006	31.936	0.000	0.193	0.220
Omnibus:	3.514	Durbin-Watson:	0.519			
Prob(Omnibus):	0.173	Jarque-Bera (JB):	2.938			
Skew:	0.861	Prob(JB):	0.230			
Kurtosis:	2.512	Cond. No.	2.09e+03			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

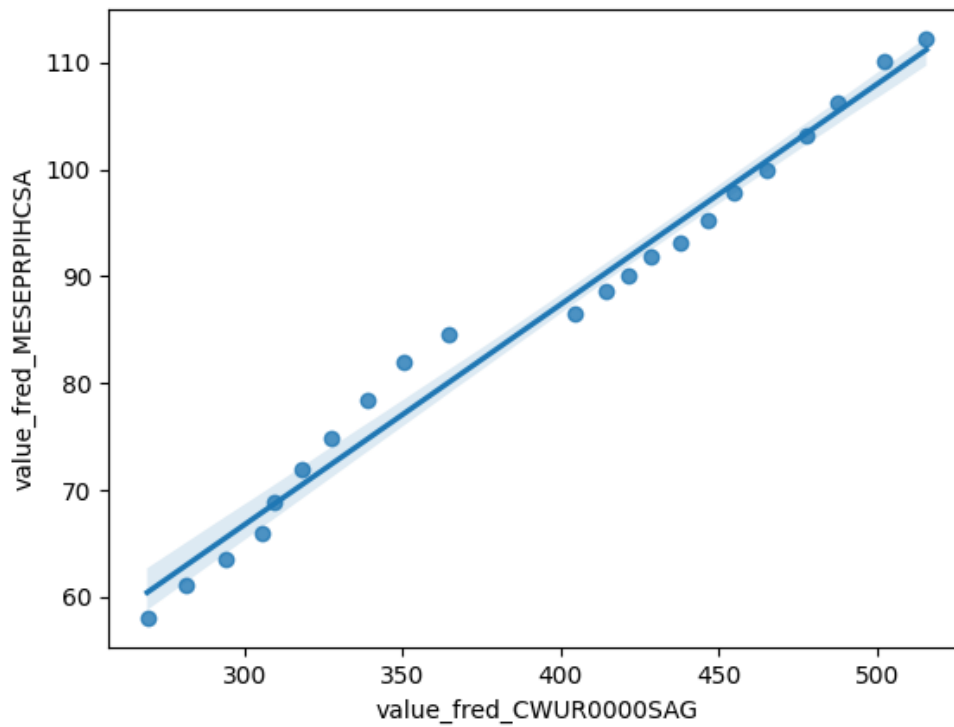


Figure 5: Raw Regression Plot for 2026-07-29

[2] The condition number is large, $2.09\text{e}+03$. This might indicate that there are strong multicollinearity or other numerical problems.

4.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_MESEPRPIHCSA_detrended	R-squared:	0.000
Model:	OLS	Adj. R-squared:	-0.050
Method:	Least Squares	F-statistic:	0.001018
Date:	Wed, 29 Jul 2026	Prob (F-statistic):	0.975
Time:	13:34:17	Log-Likelihood:	-38.006
No. Observations:	22	AIC:	80.01
Df Residuals:	20	BIC:	82.19
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.002e-12	0.304	3.29e-12	1.000	-0.635	0.635
value_fred_CWUR0000SAG_detrended	-0.0012	0.037	-0.032	0.975	-0.078	0.076

Omnibus:	2.786	Durbin-Watson:	0.239
Prob(Omnibus):	0.248	Jarque-Bera (JB):	1.903
Skew:	0.527	Prob(JB):	0.386
Kurtosis:	2.018	Cond. No.	8.25

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

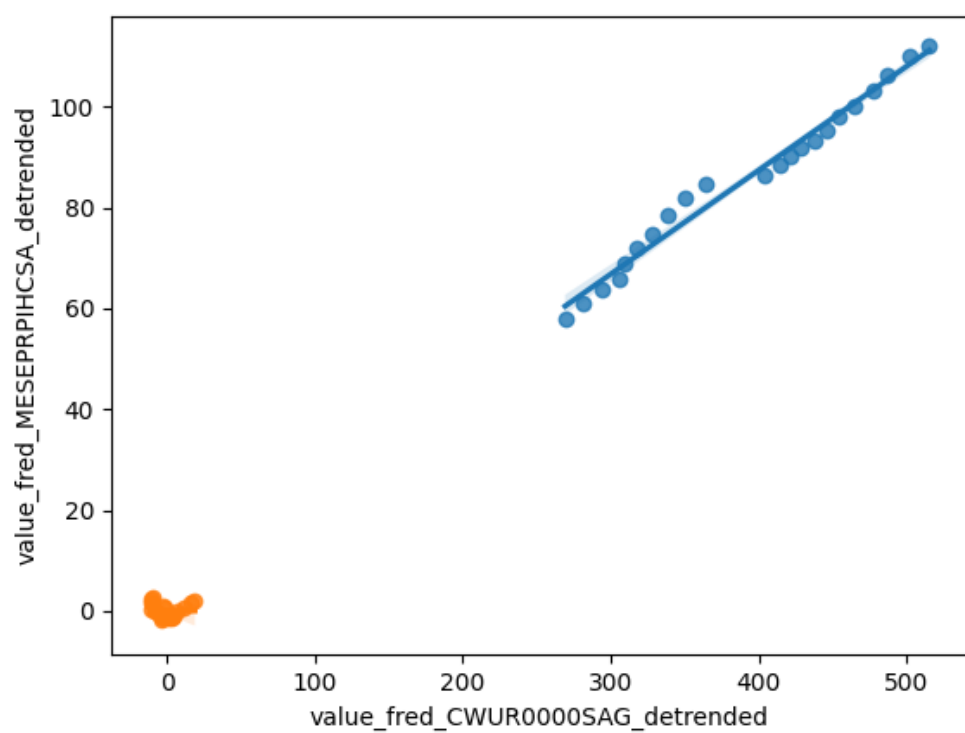


Figure 6: Detrended Regression Plot for 2026-07-29

5 Date: 2026-07-30

Series ID: B1604C1A027NBEA

This series is titled Government social benefits: to persons: State and local: Employment and training and has a frequency of Annual. The units are Billions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1974-01-01 and the observation end date is 2024-01-01. The popularity of this series is 1.

Series ID: EEZOXSLNQ

This series is titled Weighted-Average Effective Loan Rate for Zero Interval, Other Risk (Acceptable), Large Domestic Banks (DISCONTINUED) and has a frequency of Quarterly, 2nd Month's 1st Full Week. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1997-04-01 and the observation end date is 2017-04-01. The popularity of this series is 1.

5.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_EEZOXSLNQ	R-squared:	0.004
Model:	OLS	Adj. R-squared:	-0.052
Method:	Least Squares	F-statistic:	0.06894
Date:	Thu, 30 Jul 2026	Prob (F-statistic):	0.796
Time:	13:21:13	Log-Likelihood:	-43.377
No. Observations:	20	AIC:	90.75
Df Residuals:	18	BIC:	92.75
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	4.8395	3.093	1.565	0.135	-1.659	11.338
value_fred_B1604C1A027NBEA	0.6756	2.573	0.263	0.796	-4.730	6.081

Omnibus:	4.642	Durbin-Watson:	0.457
Prob(Omnibus):	0.098	Jarque-Bera (JB):	2.646
Skew:	0.672	Prob(JB):	0.266
Kurtosis:	1.830	Cond. No.	12.5

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

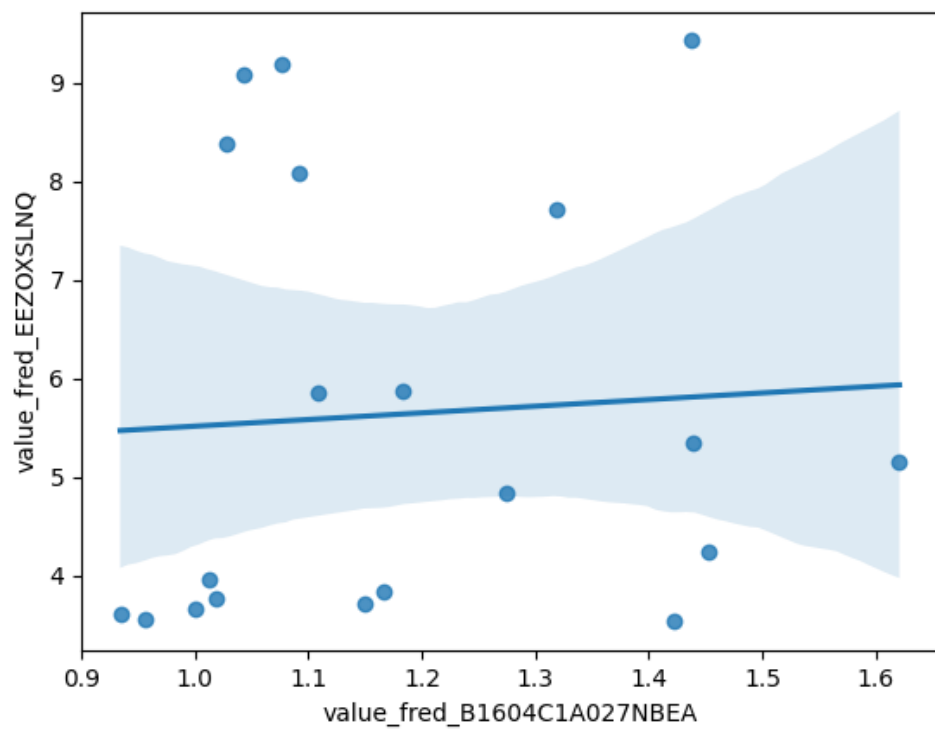


Figure 7: Raw Regression Plot for 2026-07-30

5.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_EEZOXSLNQ_detrended	R-squared:	0.217
Model:	OLS	Adj. R-squared:	0.174
Method:	Least Squares	F-statistic:	5.003
Date:	Thu, 30 Jul 2026	Prob (F-statistic):	0.0382
Time:	13:21:13	Log-Likelihood:	-30.359
No. Observations:	20	AIC:	64.72
Df Residuals:	18	BIC:	66.71
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	3.395e-13	0.260	1.3e-12	1.000	-0.547	0.547
value_fred_B1604C1A027NBEA_detrended	-3.2585	1.457	-2.237	0.038	-6.319	-0.198

Omnibus:	3.290	Durbin-Watson:	1.383
Prob(Omnibus):	0.193	Jarque-Bera (JB):	2.475
Skew:	0.851	Prob(JB):	0.290
Kurtosis:	2.731	Cond. No.	5.60

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

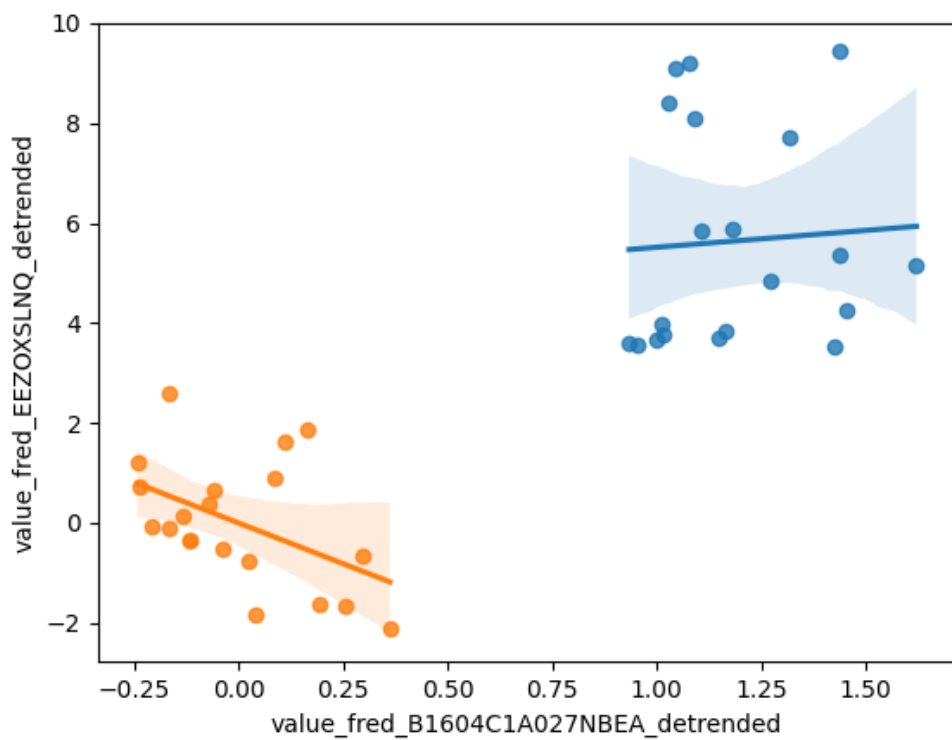


Figure 8: Detrended Regression Plot for 2026-07-30

6 Date: 2026-07-31

Series ID: GLAKPCEPCFININS

This series is titled Per Capita Personal Consumption Expenditures: Services: Financial Services and Insurance for Great Lakes BEA Region and has a frequency of Annual. The units are Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1997-01-01 and the observation end date is 2024-01-01. The popularity of this series is 1.

Series ID: DICISYPIMEPS

This series is titled Medical Services Expenditures by Disease: Diseases of the Circulatory System Price Index, MEPS Account Basis and has a frequency of Annual. The units are Index 2017=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2000-01-01 and the observation end date is 2021-01-01. The popularity of this series is 2.

6.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_DICISYPIMEPS	R-squared:	0.455
Model:	OLS	Adj. R-squared:	0.428
Method:	Least Squares	F-statistic:	16.70
Date:	Fri, 31 Jul 2026	Prob (F-statistic):	0.000574
Time:	13:24:03	Log-Likelihood:	-68.826
No. Observations:	22	AIC:	141.7
Df Residuals:	20	BIC:	143.8
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	79.4286	5.232	15.181	0.000	68.515	90.343
value_fred_GLAKPCEPCFININS	0.0083	0.002	4.087	0.001	0.004	0.013

Omnibus:	0.906	Durbin-Watson:	1.079
Prob(Omnibus):	0.636	Jarque-Bera (JB):	0.253
Skew:	0.253	Prob(JB):	0.881
Kurtosis:	3.142	Cond. No.	1.09e+04

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.09e+04. This might indicate that there are strong multicollinearity or other numerical problems.

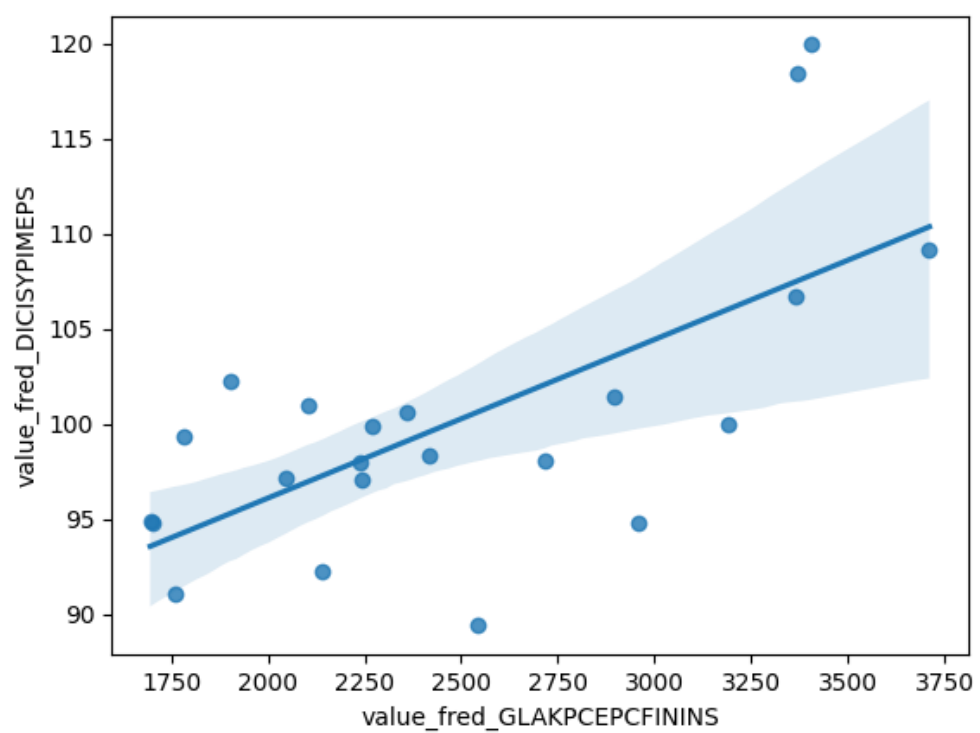


Figure 9: Raw Regression Plot for 2026-07-31

6.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_DICISYPIMEPS_detrended	R-squared:	0.077
Model:	OLS	Adj. R-squared:	0.030
Method:	Least Squares	F-statistic:	1.657
Date:	Fri, 31 Jul 2026	Prob (F-statistic):	0.213
Time:	13:24:03	Log-Likelihood:	-68.753
No. Observations:	22	AIC:	141.5
Df Residuals:	20	BIC:	143.7
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-2.898e-12	1.232	-2.35e-12	1.000	-2.569	2.569
value_fred_GLAKPCEPCFININS_detrended	0.0115	0.009	1.287	0.213	-0.007	0.030

Omnibus:	0.853	Durbin-Watson:	1.130
Prob(Omnibus):	0.653	Jarque-Bera (JB):	0.357
Skew:	0.312	Prob(JB):	0.836
Kurtosis:	2.997	Cond. No.	138.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

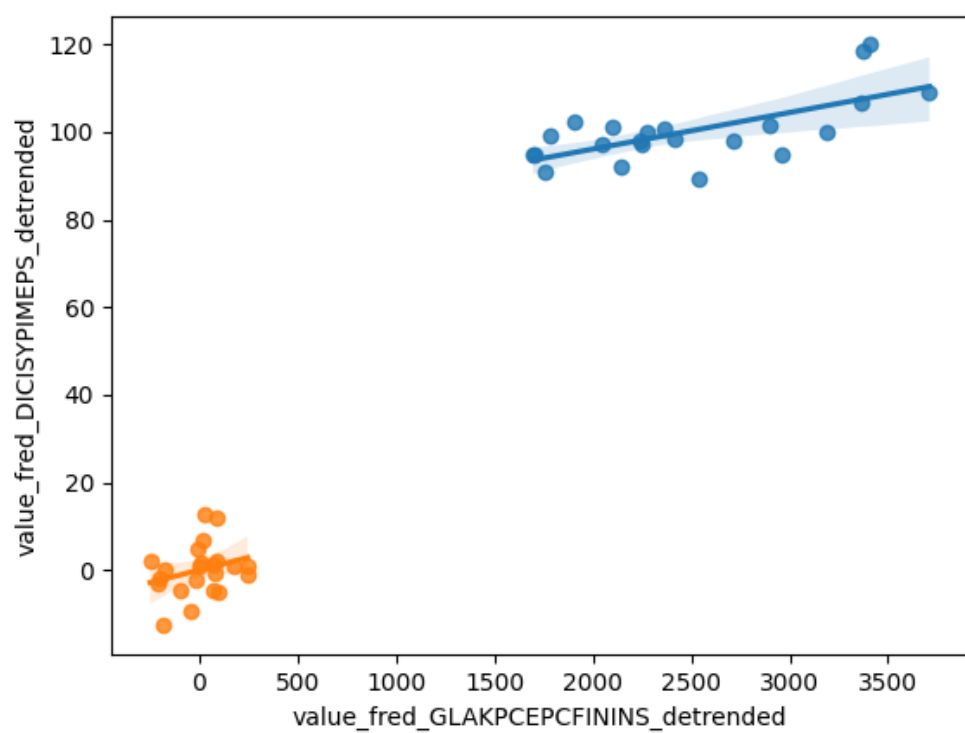


Figure 10: Detrended Regression Plot for 2026-07-31

7 Date: 2026-08-01

Series ID: WPU055121012

This series is titled Producer Price Index by Commodity: Fuels and Related Products and Power: Residential Natural Gas for Middle Atlantic Census Division and has a frequency of Monthly. The units are Index Jun 2012=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2012-06-01 and the observation end date is 2026-06-01. The popularity of this series is 1.

Series ID: SFTPGRM157SFRBSF

This series is titled San Francisco Tech Pulse (DISCONTINUED) and has a frequency of Monthly. The units are Percent Change and the seasonal adjustment is Seasonally Adjusted. The observation start date is 1971-05-01 and the observation end date is 2020-03-01. The popularity of this series is 2.

7.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_SFTPGRM157SFRBSF	R-squared:	0.068			
Model:	OLS	Adj. R-squared:	0.058			
Method:	Least Squares	F-statistic:	6.740			
Date:	Sat, 01 Aug 2026	Prob (F-statistic):	0.0110			
Time:	12:50:26	Log-Likelihood:	350.89			
No. Observations:	94	AIC:	-697.8			
Df Residuals:	92	BIC:	-692.7			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	0.0244	0.009	2.844	0.005	0.007	0.041
value_fred_WPU055121012	-0.0002	8.89e-05	-2.596	0.011	-0.000	-5.42e-05
Omnibus:	3.480	Durbin-Watson:	0.245			
Prob(Omnibus):	0.176	Jarque-Bera (JB):	3.480			
Skew:	-0.444	Prob(JB):	0.176			
Kurtosis:	2.685	Cond. No.	1.37e+03			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

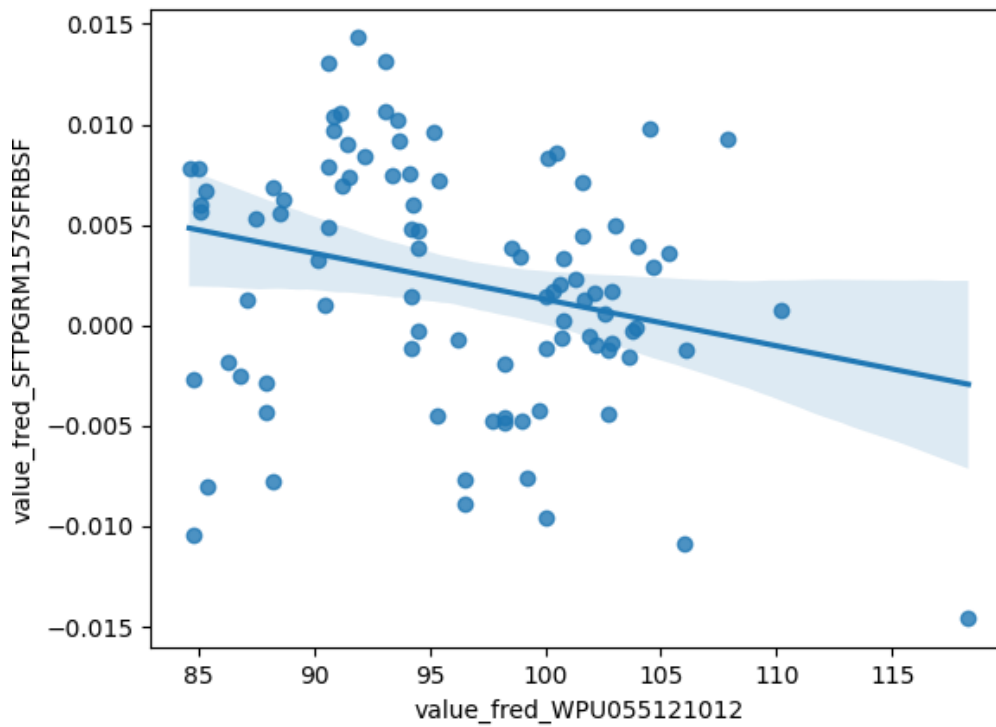


Figure 11: Raw Regression Plot for 2026-08-01

[2] The condition number is large, $1.37\text{e}+03$. This might indicate that there are strong multicollinearity or other numerical problems.

7.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_SFTPGRM157SFRBSF_detrended	R-squared:	0.043
Model:	OLS	Adj. R-squared:	0.032
Method:	Least Squares	F-statistic:	4.107
Date:	Sat, 01 Aug 2026	Prob (F-statistic):	0.0456
Time:	12:50:27	Log-Likelihood:	351.84
No. Observations:	94	AIC:	-699.7
Df Residuals:	92	BIC:	-694.6
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.069e-15	0.001	1.79e-12	1.000	-0.001	0.001
value_fred_WPU055121012_detrended	-0.0002	9.32e-05	-2.026	0.046	-0.000	-3.76e-06

Omnibus:	3.509	Durbin-Watson:	0.247
Prob(Omnibus):	0.173	Jarque-Bera (JB):	3.209
Skew:	-0.375	Prob(JB):	0.201
Kurtosis:	2.492	Cond. No.	6.41

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

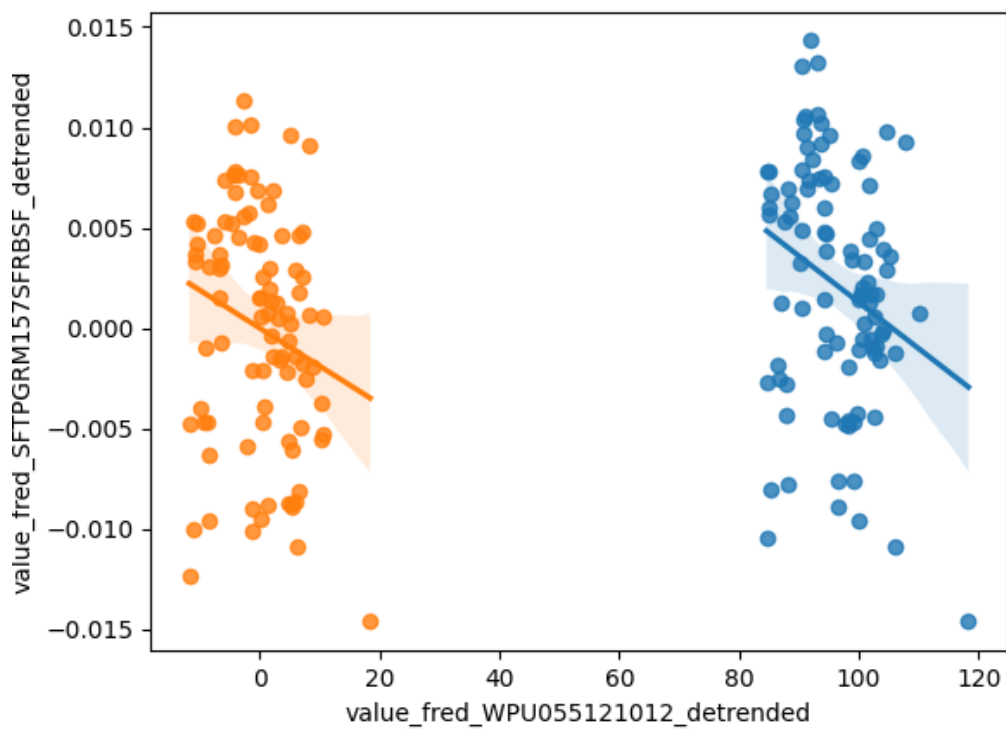


Figure 12: Detrended Regression Plot for 2026-08-01